8 **(a)** **(i)** Complete the definition of an 'electric field'.

An electric field is a region where .....

experiences ..... .

[2]

(ii) Figure 8.1 shows two parallel, oppositely charged metal plates.

Figure 8.1

On Figure 8.1, draw **four** field lines to show the pattern and the direction of the electric field between the two parallel plates. [2]

(b) Figure 8.2 shows an electric circuit that includes a thermistor, component X and a relay. The relay operates a lamp.

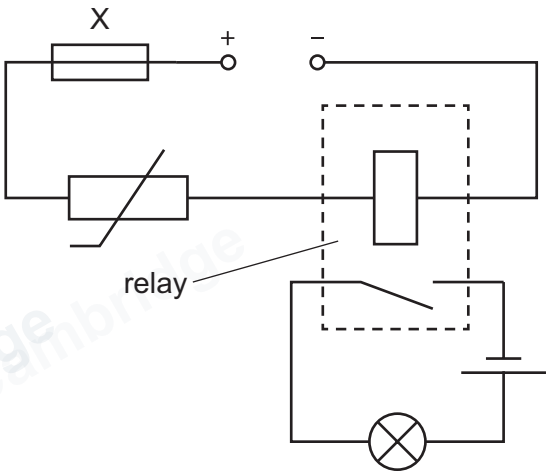

Figure 8.2

(i) Identify component X.

..... [1]

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(ii) The temperature increases. The lamp lights.

Complete the sentences to explain, in terms of potential difference (p.d.) and resistance, why the lamp lights.

When the temperature of the thermistor increases, the .....

of the thermistor ..... . This causes the p.d. across the

..... to increase. The magnetic field produced by the relay

coil ..... . The relay coil attracts ..... ,

which causes the lamp to light.

[3]

(iii) The current in the lamp is 3.2A.

Calculate the charge that flows through the lamp in 120s.

charge = ... C [2]

[Total: 10]

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